Theoretical Probability

1 Consider this list of numbers:

$$2, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 7, 7, 7, 7, 9, 9$$

a How many numbers are in the list?

b A number is chosen from the list at random. Find the probability that it is an odd number.

c State the number that has the same probability of being picked as 4.

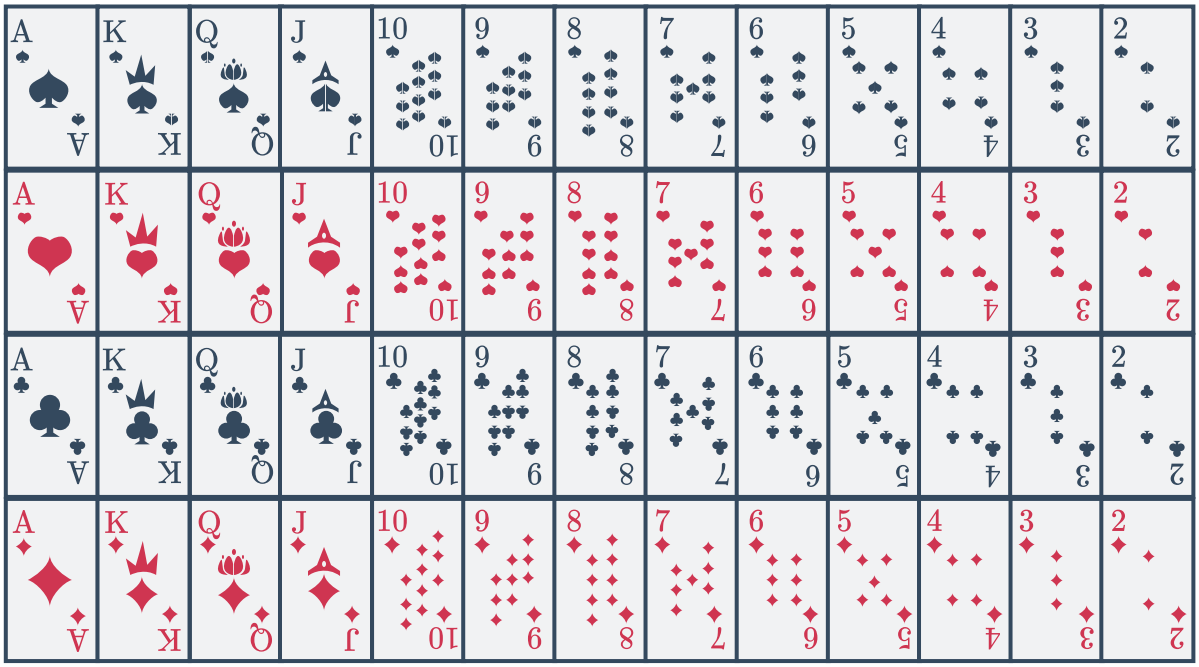
d A number is chosen from the list at random. State the number that has the highest probability of being chosen.

2 A number between 1 and 20 inclusive is randomly picked. Find:

a The probability that the number is at least 8.

b The probability that the number is less than 8.

3 A standard deck of 52 cards is shown below:



If a card is selected at random, find:

a The probability that it is a red card.

b The probability that it is a card between 5 and 9 inclusive.

c The probability that it is a card that is red and has a number between 5 and 9 inclusive.

d The probability that it is a card that is red or a king.

e The probability that it is a spade, with a number printed on it.

f The probability that it is a red card that has a number more than 2 and at most 9

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g The probability that it is a black or Kings, but not both.

4 A spinner has 4 sectors with a star, 4 sectors with apple, and 2 sectors with an elephant. All the sectors are the same size.

a Find the probability of spinning an elephant.

b Find the probability of spinning an apple.

c Are the two events in parts (a) and (b) mutually exclusive?

d Are the two events in parts (a) and (b) complementary?

Probability of complimentary events

5 The probability that Victoria will win the major prize in a raffle at the school fete is 0.053. Find the probability that Victoria will not win.

6 The probability of the local football team winning their grand final is 0.36. Find the probability that they won't win the grand final.

7 The probability that it hails today is 0.43. Find the probability that it doesn't hail.

8 A biased coin is flipped, with heads and tails as possible outcomes. Calculate $P(\text{heads})$ if $P(\text{tails}) = 0.56$.

9 A company that makes sprockets guarantees that they will be within 0.5 mm either way of the client's chosen size. There is a probability of 0.968 that a sprocket will be within the allowable size, find the probability that a sprocket won't be within the allowable size.

10 A number between 1 and 100 inclusive is randomly picked. The probability that the number is less than 61 is $\frac{60}{100}$.

a What is the complement of drawing a number greater than 61?

b Find the probability that the number is greater than 61.

11 A number between 1 and 50 inclusive is randomly picked. The probability that the number is less than 42 is $\frac{82}{100}$.

a State the complement of drawing a number less than 42.

b State the complement of drawing a number that is at least 43.

c Find the probability that the number is at least 42.

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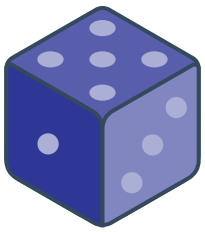
12 The 26 letters of the alphabet are written on pieces of paper and placed in a bag. If one letter is picked out of the bag at random, find the probability of:

- **a** Not selecting a B.
- **b** Not selecting a C.
- **c** Not selecting a K, R or T.
- **d** Not selecting a K, L or M.
- **e** Selecting a letter that is not in the word PROBABILITY.
- **f** Not selecting a T, L, Q, A, K or Z.
- **g** Not selecting a A, E, I, O or U.
- **h** Selecting a letter that is not in the word WORKBOOK.

13 A letter is chosen at random from the word ORDERED.

- **a** State the letter that has the highest probability of not being chosen.
- **b** Find the probability that the chosen letter is not "D".

14 A standard 6-sided die is rolled once.



- **a** Find the probability of rolling a prime number.
- **b** Find the probability of rolling an odd number.
- **c** Are the two events in parts (a) and (b) mutually exclusive?
- **d** Are the two events in parts (a) and (b) complementary?
- **e** The probability of not rolling a 2.
- **f** The probability of not rolling a 2 or 5.
- **g** The probability of not rolling an odd number.
- **h** The probability of not rolling a 9.
- **i** The probability of not rolling a 1, 2, 3, 4, 5 or 6.

15

- **a** A bag contains red marbles and blue marbles. Given the probability of drawing a red marble is $21/44$, find the probability of drawing a blue marble.

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b Another bag contains 34 red marbles and 35 blue marbles. If a marble is picked at random, find:

i $P(\text{red})$ **ii** $P(\text{Not Red})$

16 A bag contains 50 black marbles, 37 orange marbles, 29 green marbles and 23 pink marbles. If a marble is selected at random, find the following probabilities, in simplest form:

a $P(\text{orange})$ **b** $P(\text{orange or pink})$

c $P(\text{not orange})$ **d** $P(\text{neither orange nor pink})$

17 Mario has a bag of marbles. It contains 3 white marbles and 5 marbles of other colours. Mario picks a marble from the bag without looking.

a Find the probability that Mario picks a white marble.

b Find the probability that Mario picks a marble that is not white.

c Are the two events in parts (a) and (b) mutually exclusive?

d Are the two events in parts (a) and (b) complementary?

18 A marble is chosen at random from a box containing 4 different colour marbles, red, blue, green and purple. The probability of selecting different colours is given in the table. Find:

a The probability of not selecting a green marble.

b The probability of not selecting a blue marble.

c The probability of selecting a red or purple marble.

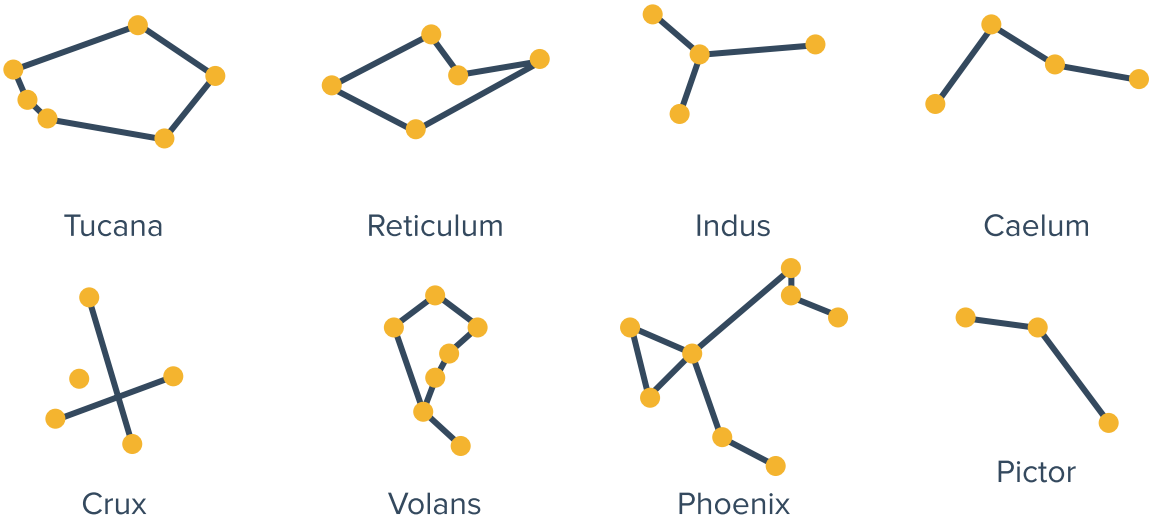
d The probability of not selecting a purple marble.

e The probability of selecting neither a blue or green marble.

Colour	Probability
Red	$\frac{3}{13}$
Blue	$\frac{2}{9}$
Green	$\frac{1}{5}$

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19 A constellation is randomly selected from the eight shown below:



a Find the probability that the name of the constellation begins with a vowel.

b The complementary event is selecting a constellation that begins with a consonant. Find the probability of this event.

c Find the probability that the constellation has 6 or more stars.

d The complementary event is selecting a constellation that has less than 6 stars. Find the probability of this event.

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